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Organocatalytic Dearomatization of β-Naphthols through a 1,6-Conjugated Addition of Alkynyl 7-Methylene-7*H*-indoles Formed In Situ

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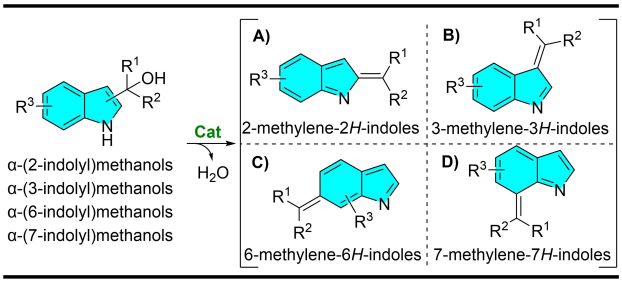
Abstract: Organocatalytic dearomatization of β-naphthols has been achieved via a 1,6-conjugated addition of 2-naphthols to alkynyl 7-methylene-7*H*-indoles *in situ* generated from α-(7-indolyl)methanols. In the presence of racemic phosphoric acid, a series of tetrasubstituted allenes was obtained in high yields. Particularly, with the aid of chiral phosphoric acid, asymmetric dearomatization of β-naphthols was investigated, affording axially chiral tetrasubstituted allenes with moderate enantioselectivity.

Keywords: Addition; Allenes; Dearomatization; Indolylmethanol; Organocatalysis

With cheap and convenient aromatic compounds as starting materials, the catalytic dearomatization reactions have paved a robust way to construct polycyclic molecules with stereogenic centers.[1] Owing to the significant advancements in asymmetric organocatalysis, notably chiral phosphoric acid catalysis,[2] organocatalytic dearomatization reactions of aromatic compounds have witnessed remarkable progress.[3] Particularly, readily available naphthols represent important platforms to construct intriguing 3D-molecular complexity.[4] It is noteworthy that organocatalytic enantioselective dearomatization reactions of naphthols have been successfully developed to incorporate stereogenic centers within the naphthyl skeletons via the formation of new carbon-carbon and hetero-carbon bonds, utilizing appropriate partners including azodicarboxylates,[5] nitroethylene,[6] 1,3-dichloro-5,5-dimethylhydantoin,[7] quinones,[8] quinone monoimides,[9] *N*-(arylthio)succinimides and *N*-(arylthio)phthalimides,[10] allenamides,[11] β,γ-alkynyl-α-imino esters,[12] and *o*-quinone diimides.[13] Importantly, these impressive reactions provide efficient access to enantioenriched naphthalenones.

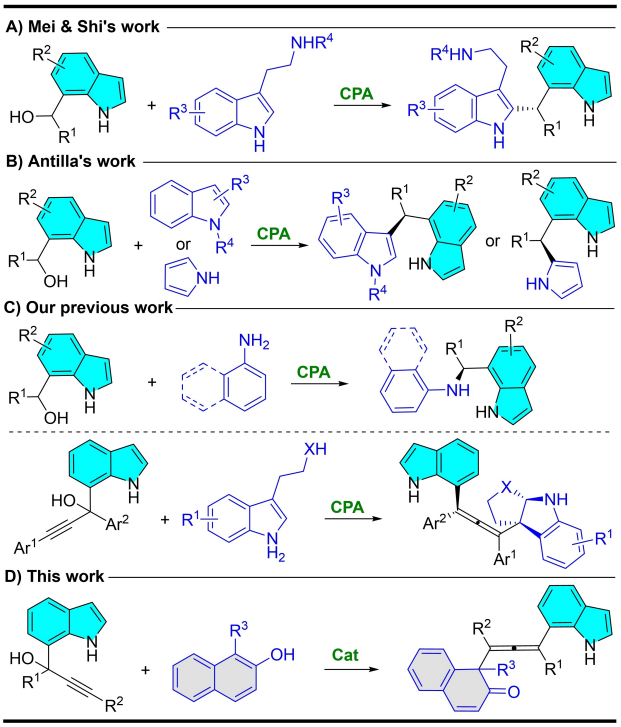
Due to its unique chemical properties, the axial chiral tetrasubstituted allene has garnered immense interest and attention from the scientific community.[14] Recently, α-indolylmethanols have emerged as versatile building blocks in organic synthesis to construct indole-based derivatives.[15] Specifically, with the aid of chiral phosphoric acids (CPAs) or their analogues, dehydration of these α-indolylmethanols in situ generated conjugate systems such as 2-methylene-2*H*-indoles (Scheme 1A),[16] 3-methylene-3*H*-indoles (Scheme 1B),[17] 6-methylene-6*H*-indoles (Scheme 1C).[18] In contrast, the organocatalytic enantioselective reactions of 7-methylene-7*H*-indoles formed in situ from α-(7-indolyl)methanols (Scheme 1D) remained underdeveloped and in an immature state. In 2018, Mei and Shi et al. reported a chiral phosphoric acid (**CPA**) catalyzed reaction of α-(7-indolyl)methanols with tryptamines for the synthesis of structurally diversified chiral 7-indolylmethanes

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Scheme 1. Conjugate systems formed in situ from the corresponding α-indolylmethanols.

(Scheme 2A).[19] Similarly, Antilla et al. also disclosed a **CPA**-catalytic formal substitution of α-(7-indolyl)methanols with aromatic partners (Scheme 2B).[18a] Notably, with aromatic amines as nucleophiles, we have established a **CPA**- catalytic aza-1,4-addition of 7-methylene-7*H*-indoles formed in situ from α-(7-indolyl)methanols (Scheme 2C).[20a] Furthermore, we also realized a **CPA**-catalyzed enantioselective 1,6-addition of tryptamines to in situ generated alkynyl 7-methylene-7*H*-indoles from the corresponding α-(7-indolyl)propargylic alcohols, affording axially chiral tetrasubstituted allenes featuring indole and hexahydropyrrolo[2,3-*b*]indole units.[20b] As a part of our continued interest in the field of organocatalytic enantioselective reactions of α-functionalized alcohols,[21] here we disclosed a phosphoric acid catalyzed dearomatization reaction of β-naphthols with alkyne 7-methylene-7*H* alcohols generated from α-(7-indolyl)propargylic alcohols (Scheme 2D). Importantly, this protocol not only achieves the dearomatization of β-naphthols but also enriches the chemistry of α-(7-indolyl)methanols.

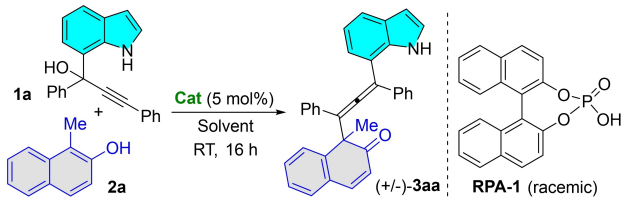


Scheme 2. Reaction of α-(7-indolyl)methanols.

The investigation began with a model reaction between α-(7-indolyl)propargylic alcohol **1 a** and 1-methyl-2-naphthol **2 a** in dichloromethane (CH_2Cl_2) at room temperature for 16 h. As shown in Table 1, trace of the desired product (+/−)-**3 aa** was obtained from the benzoic acid (**$PhCO_2H$**) catalyzed reaction (Table 1, entry 1). To our delight, the *p*-toluenesulfonic acid (**TsOH**) catalyzed reaction afforded the product (+/−)-**3 aa** in 46% yield (Table 1, entry 2). Notably, the product (+/−)-**3 aa** was obtained in 58% yield when racemic **RPA-1** was employed as catalyst (Table 1, entry 3). The next screening of solvents (Table 1, entries 4–11) identified that trichloromethane was suitable reaction media and enabled the formation of the product (+/−)-**3 aa** in 86% yield (Table 1, entry 5).

Further modification of the reaction parameters, including reaction temperature (Table 1, entry 12), catalyst loading (Table 1, entry 13), and reactant ratio (Table 1, entry 14), did not afforded better results.

Table 1. Condition optimization.



Entry[a]	Cat	Solvent	Yield [%][b]	dr[c]
1	**$PhCO_2H$**	CH_2Cl_2	< 5	> 20:1
2	**TsOH**	CH_2Cl_2	46	> 20:1
3	**RPA-1**	CH_2Cl_2	58	> 20:1
4	**RPA-1**	CH_2ClCH_2Cl	53	> 20:1
5	**RPA-1**	$CHCl_3$	86	> 20:1
6	**RPA-1**	CCl_4	53	> 20:1
7	**RPA-1**	THF	< 5	> 20:1
8	**RPA-1**	toluene	49	> 20:1
9	**RPA-1**	$PhCF_3$	54	> 20:1
10	**RPA-1**	anisole	47	> 20:1
11	**RPA-1**	PhCl	51	> 20:1
12	**RPA-1**	$CHCl_3$ (50 °C)	62	> 20:1
13	**RPA-1** (2 mol%)	$CHCl_3$	60	> 20:1
14[d]	**RPA-1**	$CHCl_3$	51	> 20:1

[a] Reaction conditions: a mixture of **1 a** (0.075 mmol), **2 a** (0.05 mmol) and **Cat** (5 mol%) in the solvent (0.3 mL) was stirred at room temperture (RT) for 16 h.
[b] Isolated yield.
[c] Determined by ^{1}H NMR analysis.
[d] **1 a** (0.05 mmol) and **2 a** (0.06 mmol) were used.

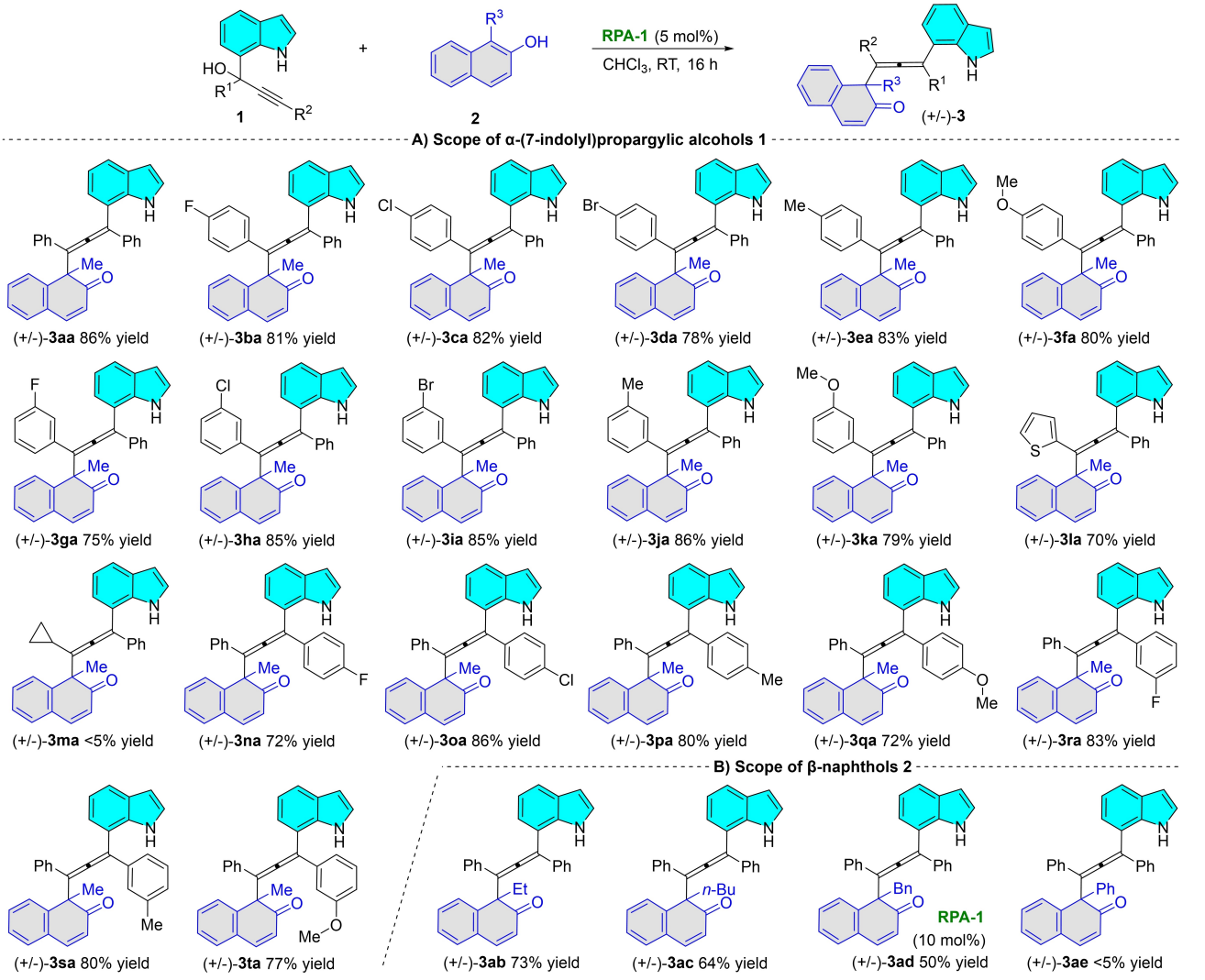
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Having the optimal conditions in hand, we examined the substrate scope of **RPA-1** catalyzed reactions (Table 2). Firstly, the range of α-(7-indolyl)propargylic alcohols **1** was investigated by **RPA-1** catalyzed reaction of 1-methyl-2-naphthol **2a** (Table 2A). In general, a broad range of α-(7-indolyl)methanols is compatible, providing the corresponding adducts in high yields. Specifically, the substituent R^1/R^2 with electron-withdrawing (F, Cl, and Br) or electron-donating (Me and MeO) groups slightly affected the yield, affording the corresponding products (+/−)-**3aa-3ka**, (+/−)-**3na-3ta** in 70–86% yields. Importantly, neither significant electronic effects nor steric hindrance effects were observed. Moreover, α-(7-indolyl)methanol **1l** with heteroaromatic substituent (R^1 = Ph, R^2 = 2-thienyl) was also tolerated, affording the product (+/−)-**3la** in a yield of 70%. Exceptionally, trace of product (+/−)-**3ma** was obtained from the **RPA-1** catalyzed reaction of α-(7-indolyl)methanol with aliphatic substituents (R^1 = Ph, R^2 = cyclopropyl). Subsequently, we investigated the substrate range of β-naphthol (Table 2B). It was observed that the substituent R^3 exerted a pronounced impact on the yield. Substituting the methyl group with alternative groups such as ethyl, *n*-butyl (*n*-Bu), and benzyl (Bn), led to suboptimal yields, affording the corresponding products (+/−)-**3ab-ad** in 50–73% yield. Particularly, 1-phenyl-2-naphthol **2e** was found to be incompatible under the established reaction conditions.

To demonstrate the usefulness of this protocol, the scale-up of the reaction was investigated. When the reaction was scaled up to 2.0 mmol under the standard conditions, the desired product of (+/−)-**3aa** was obtained in 83% yield (Scheme 3A). In addition,

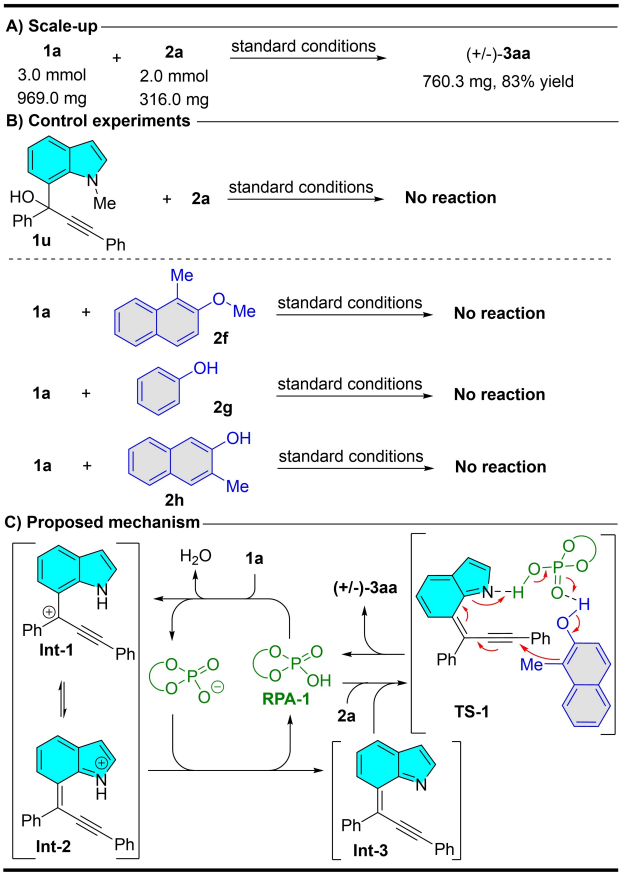
Table 2. Substrate scope of the **RPA-1** catalyzed reactions.[a]



[a] A mixture of **1** (0.075 mmol), **2** (0.05 mmol), **RPA-1** (5 mol%) in $CHCl_3$ (0.3 mL) was stirred at RT for 16 h. All dr > 20:1, determined by ^{1}H NMR analysis.

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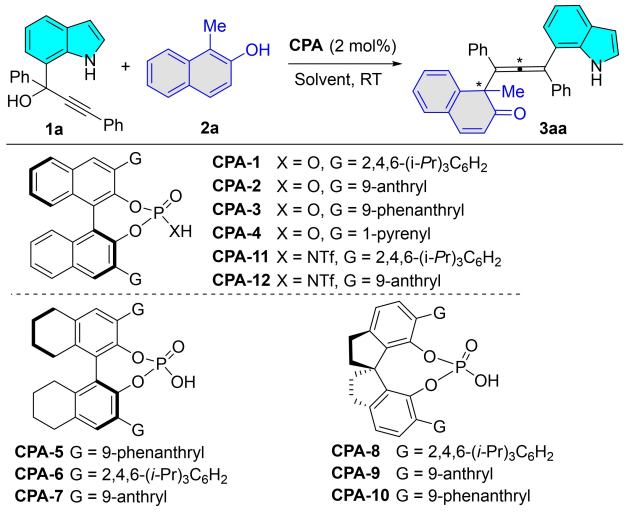


Scheme 3. Further investigations and proposed mechanism.

controlled experiments were performed to gain insight into the reaction mechanism (Scheme 3B). No reaction between α-(7-indolyl)propargylic alcohol **1u** and β-naphthol **2a** occurred, which indicated that the free N–H group of α-(7-indolyl)methanol **1a** plays a crucial role in controlling efficiency. When the hydroxyl group of 1-methyl-2-naphthol **2a** was masked with methyl group, no desired product was obtained, disclosing that the free O–H group of β-naphthol **2a** also plays a decisive role in the occurrence of this reaction. Furthermore, both phenol **2g** and 3-methyl-2-naphthol **2h** were found to be not compatible under the standard conditions, which indicated the limitation of the reaction. Accordingly, a possible reaction mechanism was proposed in Scheme 1C. In the presence of catalyst **RPA-1**, dehydration of α-(7-indolyl)propargylic alcohol **1a** generate propargyl carbocation **Int-1**, which could easily isomerize into intermediate **Int-2**. After deprotonation, the key intermediate alkynyl 7-methylene-7*H*-indole **Int-3** was formed and followed by asymmetric 1,6-conjugate addition of 1-methyl-2-naphthol **2a** via transition state **TS-1** to give the desired product **(+/−)-3aa**. Importantly, the "double-hydrogen bond" activation model determines the diastereoselectivity.

In addition, we performed a preliminary survey of the organocatalytic enantioselective version of this 1,6-addition reaction (Table 3). A series of BINOL-derived chiral phosphoric acids (**CPA**s) were examined via the

Table 3. Investigations on **CPA**-catalyzed reactions.[a]



Entry	Cat	Solvent	Yield[%][b]	ee[%][c]	dr[d]
1	**CPA-1**	CH_2Cl_2	60	17	>20:1
2	**CPA-2**	CH_2Cl_2	69	43	>20:1
3	**CPA-3**	CH_2Cl_2	76	35	>20:1
4	**CPA-4**	CH_2Cl_2	73	41	>20:1
5	**CPA-5**	CH_2Cl_2	72	41	>20:1
6	**CPA-6**	CH_2Cl_2	60	19	>20:1
7	**CPA-7**	CH_2Cl_2	72	41	>20:1
8	**CPA-8**	CH_2Cl_2	39	28	>20:1
9	**CPA-9**	CH_2Cl_2	75	18	>20:1
10	**CPA-10**	CH_2Cl_2	68	11	>20:1
11	**CPA-11**	CH_2Cl_2	79	17	>20:1
12	**CPA-12**	CH_2Cl_2	57	43	>20:1
13	**CPA-2**	$CHCl_3$	68	55	>20:1
14	**CPA-2**	CCl_4	67	58	>20:1
15	**CPA-2**	toluene	80	60	>20:1
16	**CPA-2**	$PhCF_3$	91	55	>20:1
17	**CPA-2**	xylenes	58	54	>20:1
18	**CPA-2**	THF	<5	-	>20:1
19	**CPA-2**	anisole	64	52	>20:1
20[e]	**CPA-2**	toluene	60	56	>20:1
21[f]	**CPA-2**	toluene	56	62	>20:1
22[g]	**CPA-2**	toluene	86	60	>20:1
23[h]	**CPA-2**	toluene	87	55	>20:1

[a] Reaction conditions: a mixture of **1a** (0.05 mmol), **2a** (0.06 mmol) and **CPA** (2 mol%) in the solvent (0.3 mL) were stirred at room temperture for 16 h.
[b] Isolated yield.
[c] Determined by HPLC analysis.
[d] Determined by ^{1}H NMR analysis.
[e] 4 Å MS (30.0 mg) was used as additive.
[f] At 0 °C, 60 h.
[g] Toluene (0.5 mL), 20 h.
[h] Toluene (1.0 mL), 30 h.

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model reaction between α-(7-indolyl)propargylic alcohol **1 a** and 1-methyl-2-naphthol **2 a** in dichloromethane at room temperature (Table 3, entries 1–12). To our delight, all these **CPA**s could catalyze the reaction to furnish chiral product. In particular, **CPA-2** enabled the formation of the desired chiral tetrasubstituted allene **3 aa** in 69% yield with 43% ee and >20:1 dr (table 3, entry 2). In order to improve the enantioselectivity, investigations on the effect of reaction media were carried out (Table 3, entries 13–19). Notably, toluene was identified as the most suitable reaction medium to obtain product **3 aa** in 80% yield and 60% ee and >20:1 dr (Table 3, entry 15). To further improve enantioselectivity, we modified the reaction parameters including additives, temperature, and concentration, and no better results were obtained (Table 3, entries 20–23). Although the enantioselectivity was moderate, these preliminary results demonstrated the possibility of constructing tetrasubstituted allene.

In summary, we developed an organocatalytic dearomatization of β-naphthols via an 1,6-conjugated addition of alkynyl 7-methylene-7*H*-indoles in situ generated from α-(7-indolyl)propargylic alcohols to synthesize axially chiral tetrasubstituted allenes. Using this method, a series of tetrasubstituted allenes were obtained in high yields. Based on this foundation, we delved into the exploration of the reaction's enantioselectivity and subsequently confirmed the practicality of achieving such enantioselectivity. We expect this study to stimulate more efforts to develop the enantioselective dearomatization of β-naphthols and enrich the chemistry of α-(7-indolyl)methanols and axially chiral tetrasubstituted allenes.

Experimental Section

To a solution of $CHCl_3$ (0.3 mL) was added α-(7-indolyl)propargylic alcohols **1** (0.075 mmol), 2-naphthols **2** (0.05 mmol), and **RPA-1** (0.0025 mmol). The reaction mixture was stirred at room temperature for 16 h, and then the solvent was removed under vacuum to give a residue, which was purified by silica gel chromatography to yield the desired products (+/−)-**3**.

Acknowledgements

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